

Lab3

October 30, 2025

```
[1]: # %pip install --upgrade pip
# %pip install "scanpy>=1.10" anndata matplotlib numpy pandas igraph leidenalg
```

```
[2]: import os
from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import scanpy as sc

sc.settings.verbosity = 3
sc.set_figure_params(facecolor="white", figsize=(6, 6))
```

```
[3]: adata = sc.read_h5ad("../docs/
↳Visium_Human_Breast_Cancer_filtered_feature_bc_matrix.h5ad")
adata.var_names_make_unique()
adata
```

```
[3]: AnnData object with n_obs × n_vars = 4898 × 36601
      obs: 'in_tissue', 'array_row', 'array_col', 'imagecol', 'imagerow'
      var: 'gene_ids', 'feature_types', 'genome'
      uns: 'spatial'
      obsm: 'spatial'
```

```
[4]: print(adata.X.shape)           # 1.
print(adata.var['gene_ids'].shape)  # 2.   ID
print(adata.obs['in_tissue'].shape) # 3.
print(adata.obsm['spatial'].shape)  # 4.
```

```
(4898, 36601)
(36601,)
(4898,)
(4898, 2)
```

```
[5]: # Flag mitochondrial genes and compute QC metrics
adata.var["mt"] = adata.var_names.str.upper().str.startswith("MT-")
sc.pp.calculate_qc_metrics(adata, qc_vars=["mt"], inplace=True)
```

```

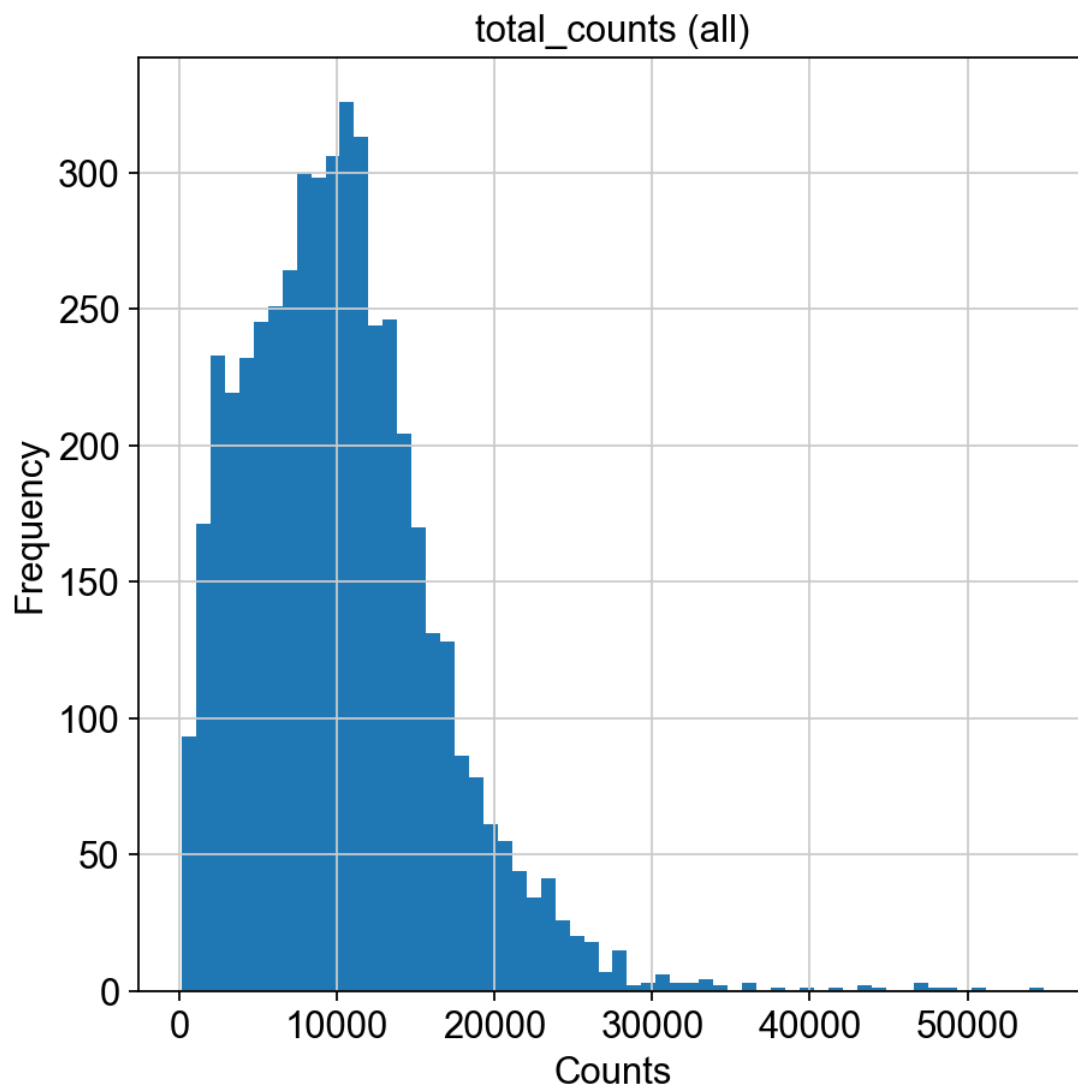
# Histogram: total_counts (all)
plt.figure()
plt.hist(adata.obs["total_counts"], bins=60)
plt.title("total_counts (all)")
plt.xlabel("Counts")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

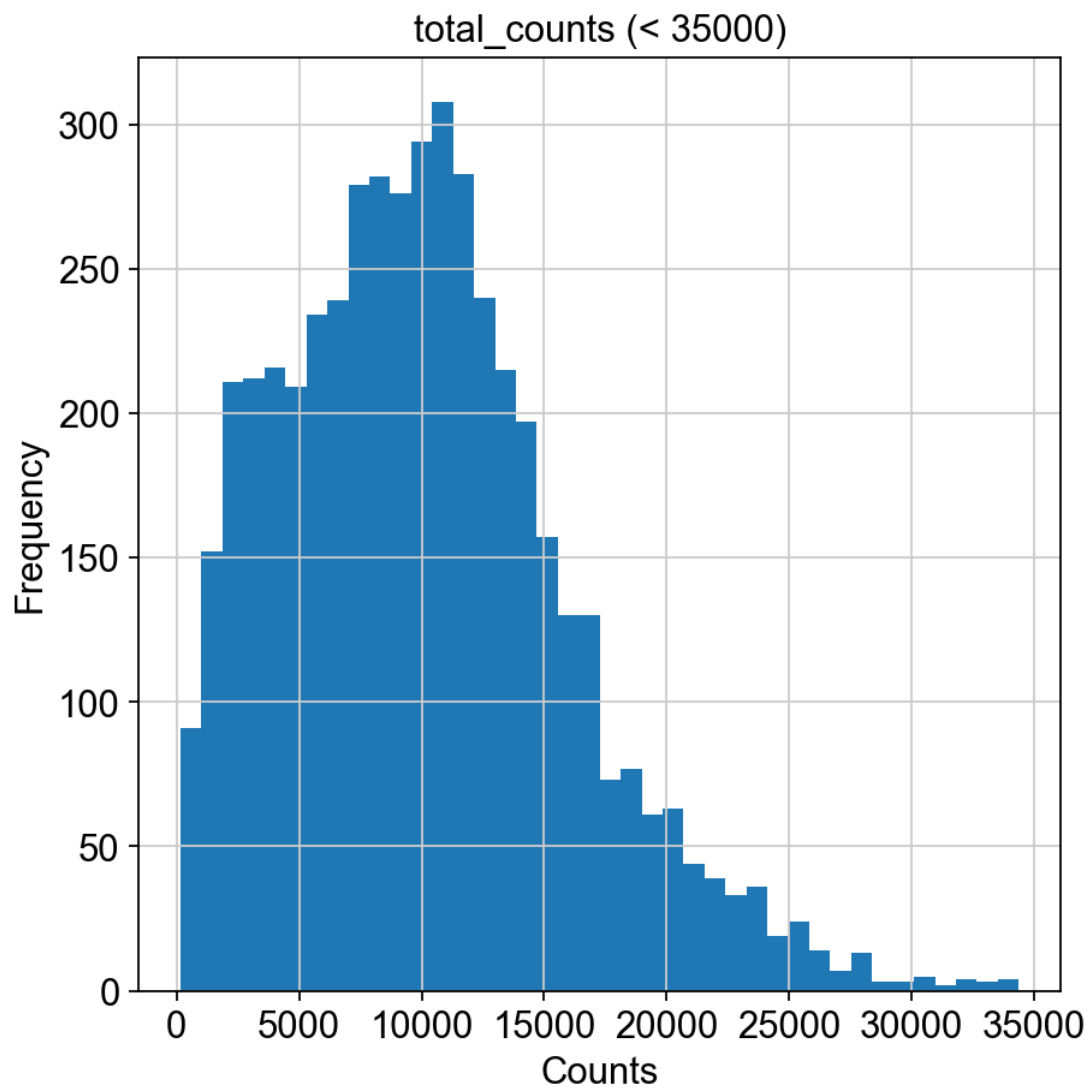
# Histogram: total_counts (< 35000)
plt.figure()
vals = adata.obs["total_counts"]
plt.hist(vals[vals < 35000], bins=40)
plt.title("total_counts (< 35000)")
plt.xlabel("Counts")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

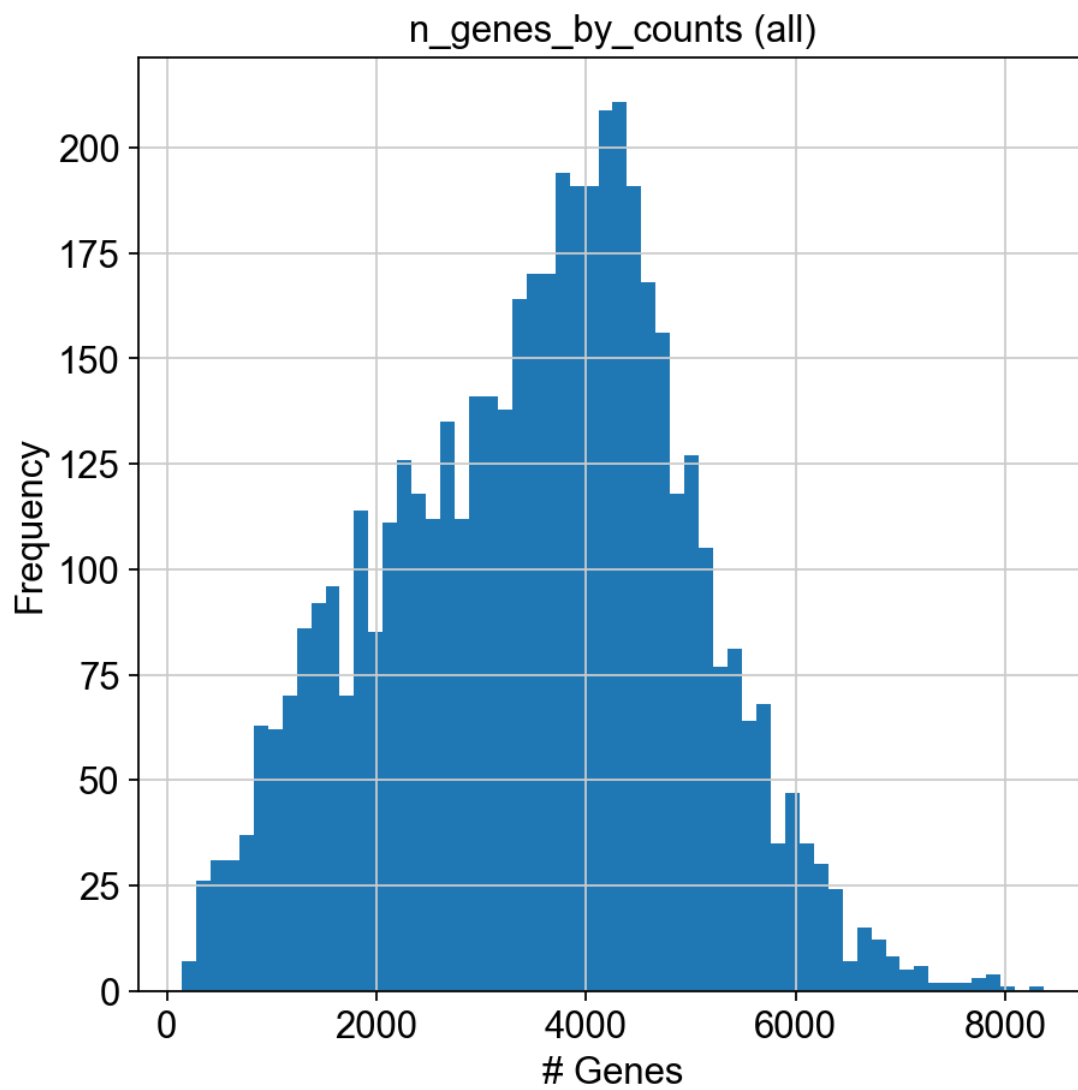
# Histogram: n_genes_by_counts (all)
plt.figure()
plt.hist(adata.obs["n_genes_by_counts"], bins=60)
plt.title("n_genes_by_counts (all)")
plt.xlabel("# Genes")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

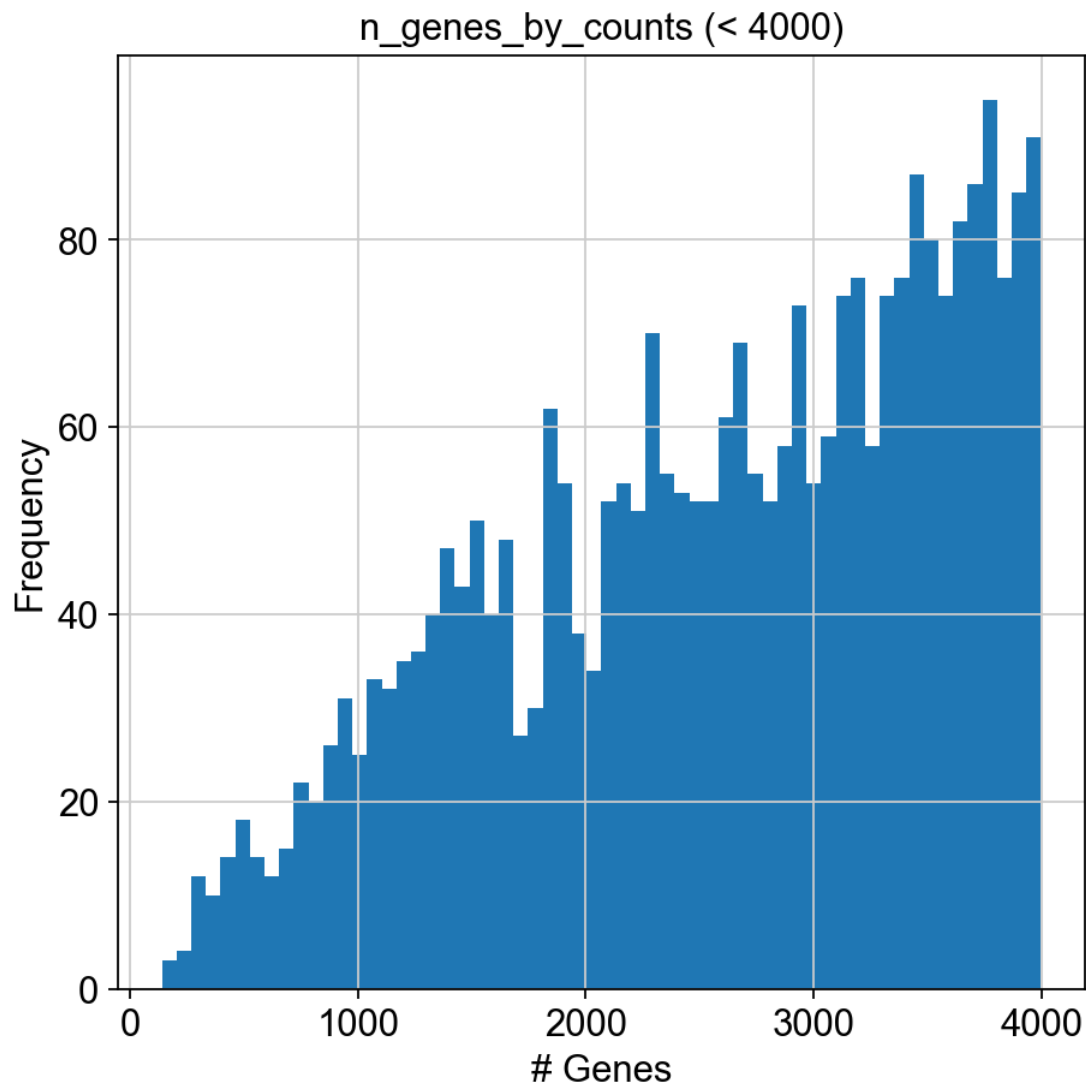
# Histogram: n_genes_by_counts (< 4000)
plt.figure()
vals2 = adata.obs["n_genes_by_counts"]
plt.hist(vals2[vals2 < 4000], bins=60)
plt.title("n_genes_by_counts (< 4000)")
plt.xlabel("# Genes")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

```









```
[6]: sc.pp.filter_cells(adata, min_counts=5000)
sc.pp.filter_cells(adata, max_counts=35000)
adata = adata[adata.obs["pct_counts_mt"] < 20].copy()
print("#spots after MT filter:", adata.n_obs)
sc.pp.filter_genes(adata, min_cells=10)
print("After gene filter, n_vars:", adata.n_vars)
```

```
filtered out 1027 cells that have less than 5000 counts
filtered out 16 cells that have more than 35000 counts
#spots after MT filter: 3855
filtered out 18143 genes that are detected in less than 10 cells
After gene filter, n_vars: 18458
```

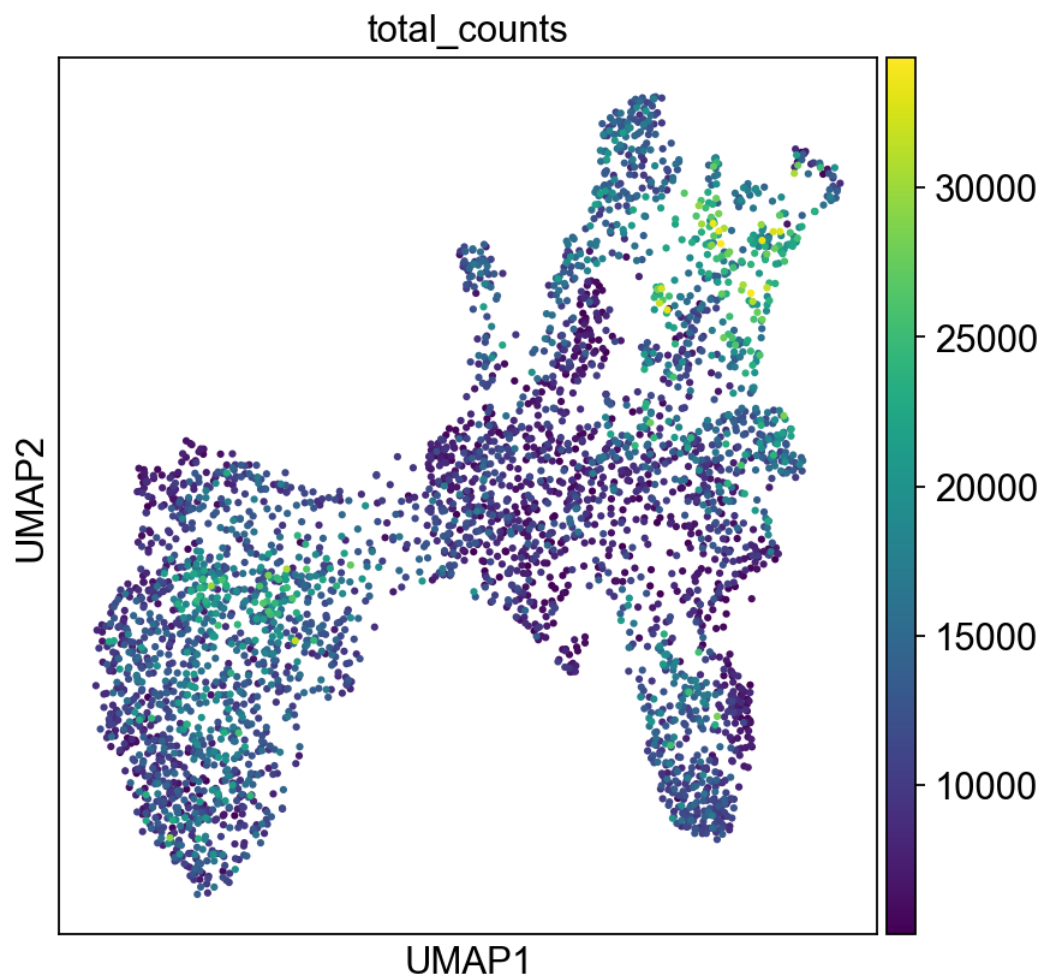
```
[7]: sc.pp.normalize_total(adata, inplace=True)
sc.pp.log1p(adata)
sc.pp.highly_variable_genes(adata, flavor="seurat", n_top_genes=2000)
print("HVGs identified:", int(adata.var["highly_variable"].sum()))
```

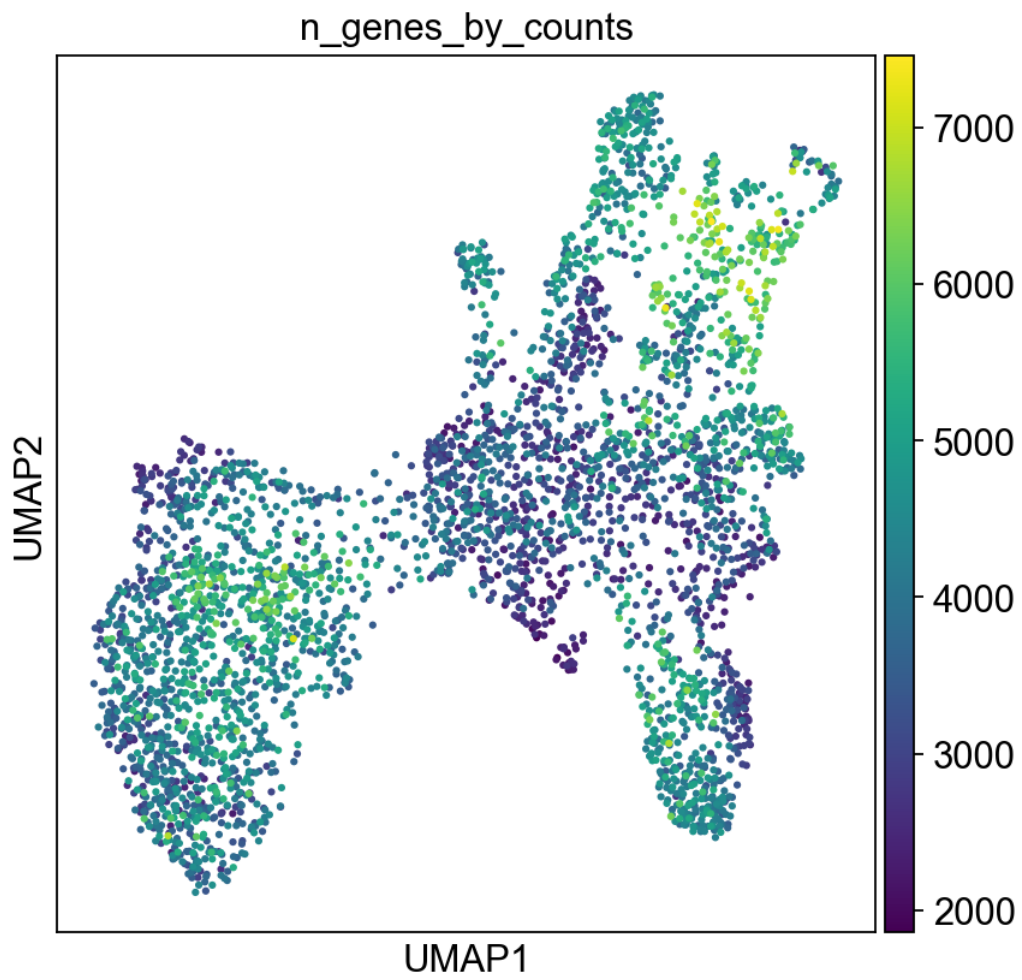
```
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
HVGs identified: 2000
```

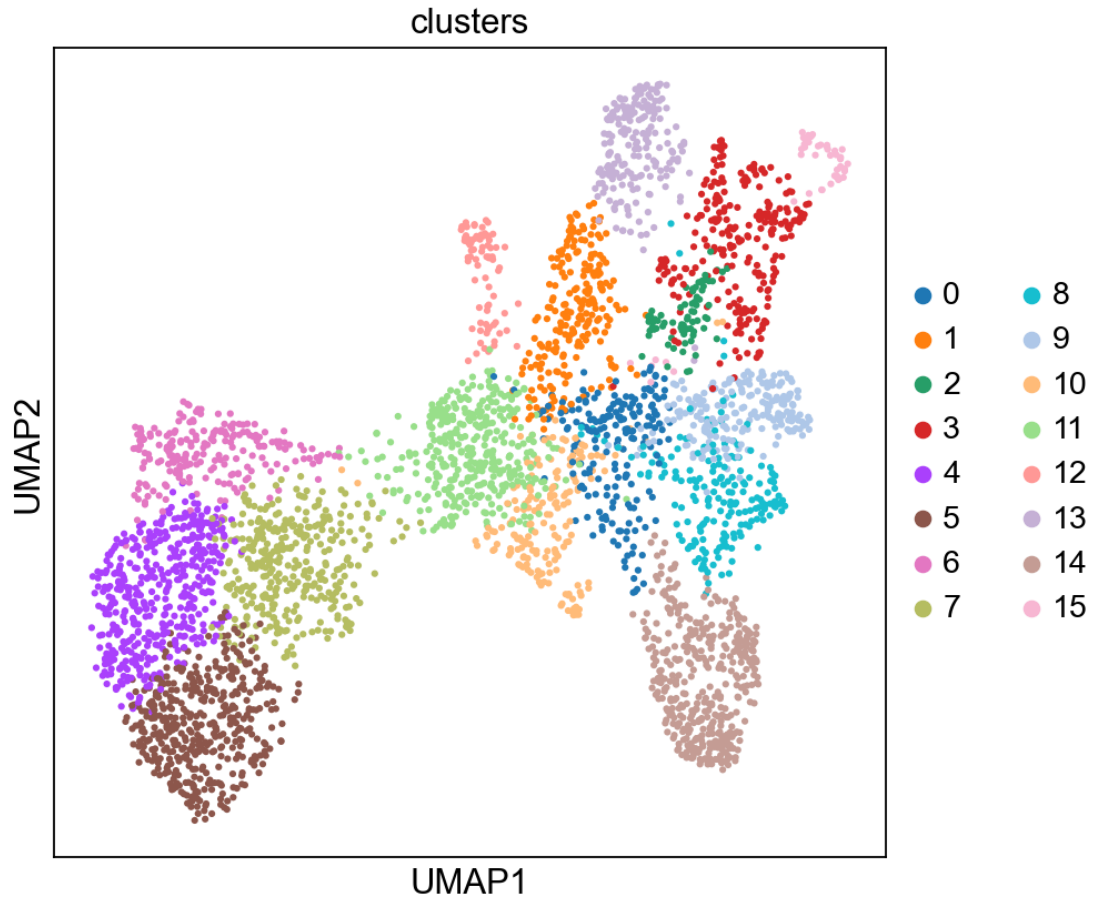
```
[8]: sc.pp.pca(adata)
sc.pp.neighbors(adata)
sc.tl.umap(adata)
sc.tl.leiden(
    adata, key_added="clusters", flavor="igraph",
    directed=False, n_iterations=2, resolution=1.0
)
print("Leiden clusters:", adata.obs["clusters"].unique().tolist())
```

```
computing PCA
  with n_comps=50
  with n_comps=50
  finished (0:00:01)
computing neighbors
  using 'X_pca' with n_pcs = 50
  finished: added to `uns['neighbors']`
  `obsp['distances']`, distances for each pair of neighbors
  `obsp['connectivities']`, weighted adjacency matrix (0:00:02)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm)
  'umap', UMAP parameters (adata.uns) (0:00:04)
running Leiden clustering
  finished: found 16 clusters and added
  'clusters', the cluster labels (adata.obs, categorical) (0:00:00)
Leiden clusters: ['0', '1', '2', '4', '6', '5', '7', '3', '8', '11', '9', '10',
'13', '14', '15', '12']
```

```
[12]: # UMAP plots
sc.pl.umap(adata, color=["total_counts"], wspace=0.4)
sc.pl.umap(adata, color=["n_genes_by_counts"], wspace=0.4)
sc.pl.umap(adata, color=["clusters"], wspace=0.4)
```

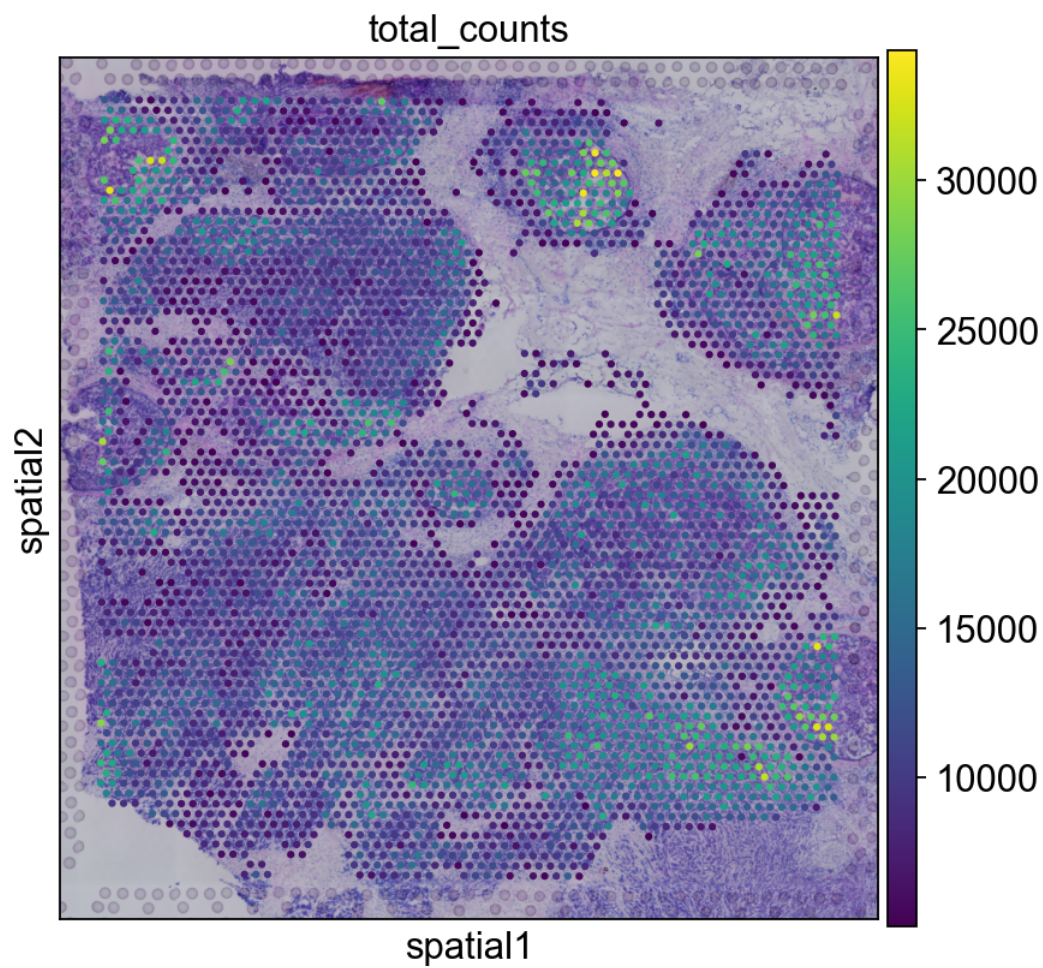




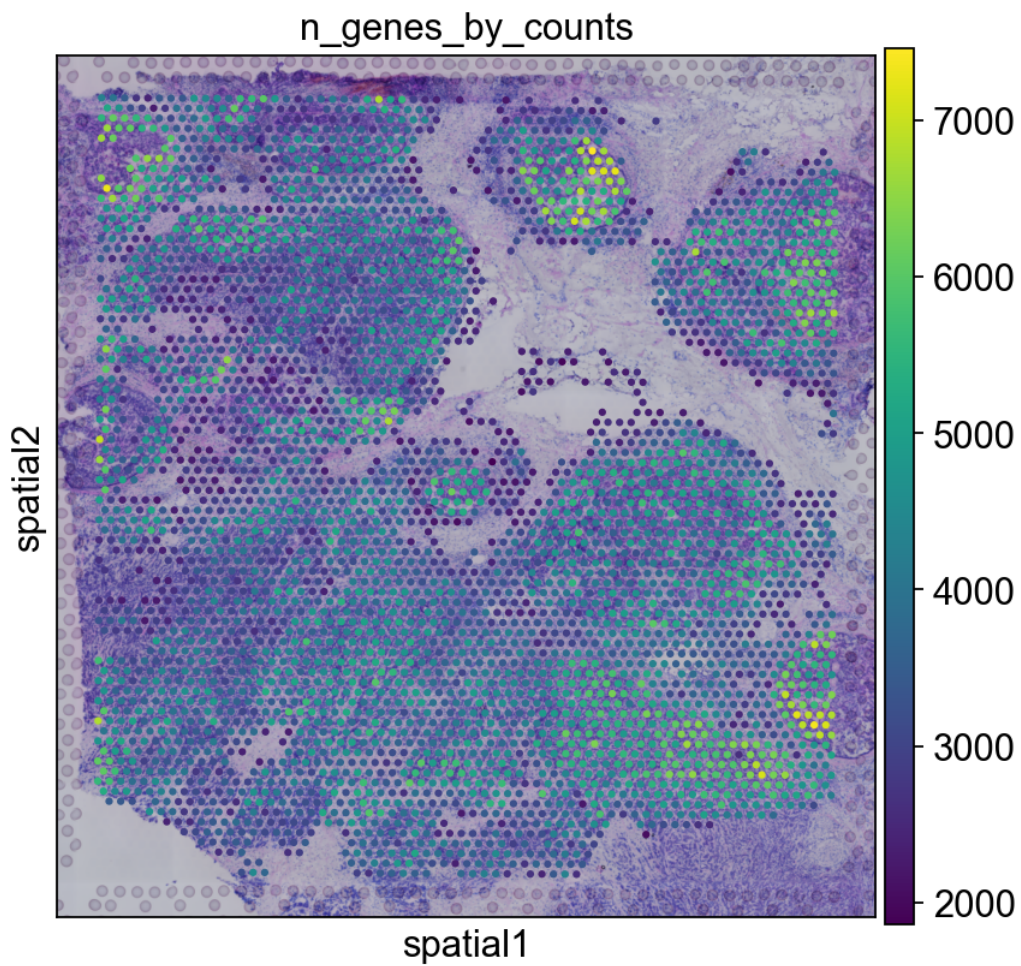


```
[13]: # Spatial QC overlays
sc.pl.spatial(adata, color=["total_counts"])
sc.pl.spatial(adata, color=["n_genes_by_counts"])
sc.pl.spatial(adata, color="clusters", size=1.5)
```

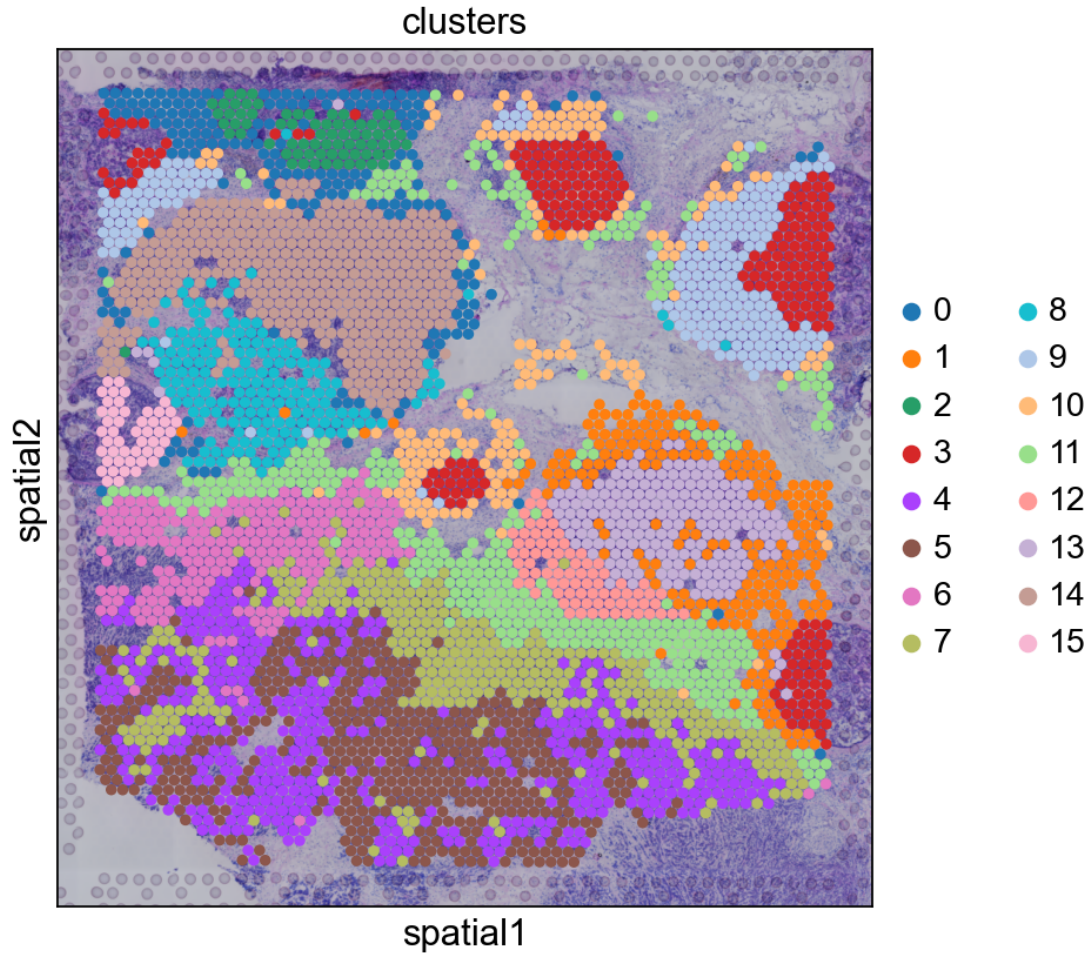
```
/var/folders/w2/jm4sr6hx73bdcs3t0f0d0x9h0000gn/T/ipykernel_51590/833416663.py:2:
FutureWarning: Use `squidpy.pl.spatial_scatter` instead.
sc.pl.spatial(adata, color=["total_counts"])
```



```
/var/folders/w2/jm4sr6hx73bdcs3t0f0d0x9h0000gn/T/ipykernel_51590/833416663.py:3:  
FutureWarning: Use `squidpy.pl.spatial_scatter` instead.  
    sc.pl.spatial(adata, color=["n_genes_by_counts"])
```



```
/var/folders/w2/jm4sr6hx73bdcs3t0f0d0x9h0000gn/T/ipykernel_51590/833416663.py:4:  
FutureWarning: Use `squidpy.pl.spatial_scatter` instead.  
    sc.pl.spatial(adata, color="clusters", size=1.5)
```

```
[11]: # Marker genes for one cluster (e.g., top genes for each cluster)
sc.tl.rank_genes_groups(adata, "clusters", method="t-test")
sc.pl.rank_genes_groups_heatmap(
    adata, groups=None, n_genes=5, groupby="clusters")
```

ranking genes

finished: added to `uns['rank\_genes\_groups']`

'names', sorted np.recarray to be indexed by group ids

'scores', sorted np.recarray to be indexed by group ids

'logfoldchanges', sorted np.recarray to be indexed by group ids

'pvals', sorted np.recarray to be indexed by group ids

'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:01)

WARNING: dendrogram data not found (using key=dendrogram_clusters). Running
`sc.tl.dendrogram` with default parameters. For fine tuning it is recommended to
run `sc.tl.dendrogram` independently.

using 'X_pca' with n_pcs = 50

Storing dendrogram info using `uns['dendrogram\_clusters']`

WARNING: Gene labels are not shown when more than 50 genes are visualized. To show gene labels set `show\_gene\_labels=True`

